

Exploring the Landscape of Natural Language Processing Research

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Abstract

As an efficient approach to understand, generate, and process natural language texts, research in natural language processing (NLP) has exhibited a rapid spread and wide adoption in recent years. Given the increasing amount of research work in this area, several NLP-related approaches have been surveyed in the research community. However, a comprehensive study that categorizes established topics, identifies trends, and outlines areas for future research remains absent to this day. Contributing to closing this gap, we have systematically classified and analyzed research papers included in the ACL Anthology. As a result, we present a structured overview of the research landscape, provide a taxonomy of fields-of-study in NLP, analyze recent developments in NLP, summarize our findings, and highlight directions for future work.¹

1 Introduction

Natural language is a fundamental aspect of human communication and inherent to human utterances and information sharing. Accordingly, most human-generated digital data are composed in natural language. Given the ever-increasing amount and importance of digital data, it is not surprising that computational linguists have started developing ideas on enabling machines to understand, generate, and process natural language since the 1950s (Hutchins, 1999).

More recently, the introduction of the transformer model (Vaswani et al., 2017) and pretrained language models (Radford and Narasimhan, 2018; Devlin et al., 2019) have sparked increasing interest in natural language processing (NLP). Submissions on various NLP topics and applications are being published in a growing number of journals

and conferences such as TACL, ACL, and EMNLP, as well as in several smaller workshops that focus on specific areas. Thereby, the ACL Anthology² as a repository for publications from many major NLP journals, conferences and workshops emerges as an important tool for researchers. As of January 2023, it provides access to over 80,000 articles published since 1952. Figure 1 shows the distribution of publications in the ACL Anthology over the 50-year observation period.

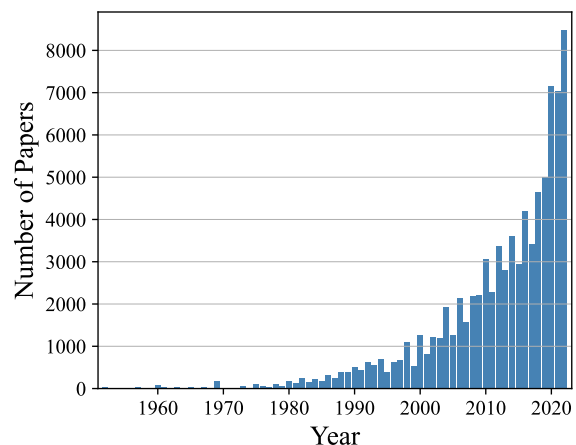


Figure 1: Distribution of number of papers per year in the ACL Anthology from 1952 to 2022.

Accompanying the increase in publications, there has also been a growth in the number of different fields-of-study (FoS) that have been researched within the NLP domain. FoS are academic disciplines and concepts that usually consist of (but are not limited to) tasks or techniques (Shen et al., 2018). Given the rapid developments in NLP research, obtaining an overview of the domain and maintaining it is difficult. As such, collecting insights, consolidating existing results, and presenting a structured overview of the field is important.

¹Code available: <https://github.com/sebischair/Exploring-NLP-Research>

²<https://aclanthology.org>

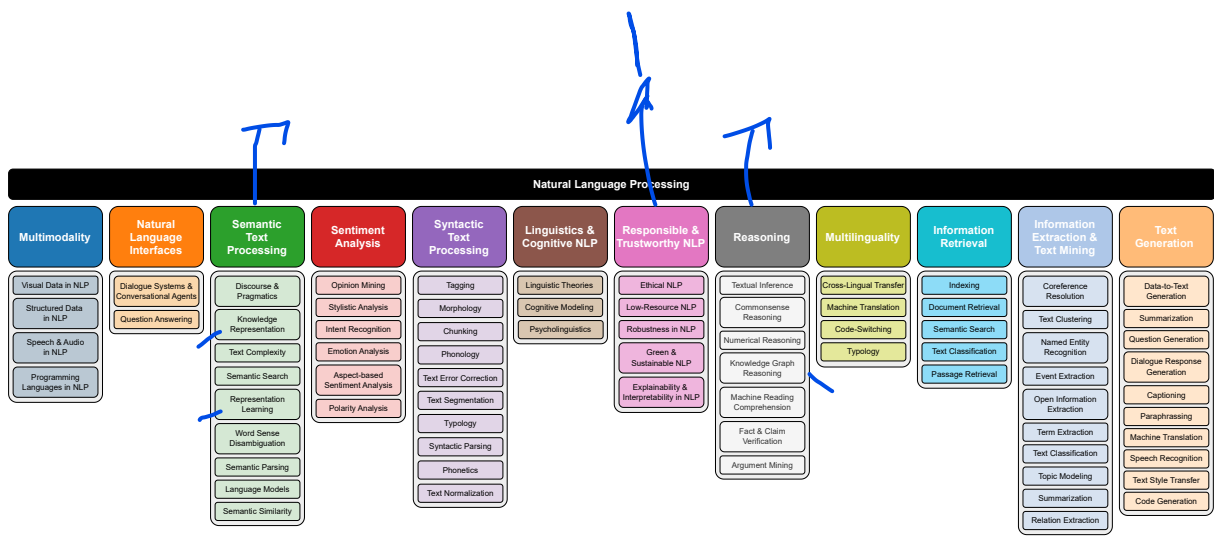


Figure 2: Taxonomy of fields-of-study in NLP. Appendix A includes more detailed descriptions of the fields-of-studies.

However, to the best of our knowledge, no studies exist yet that offer an overview of the entire landscape of NLP research. To bridge this gap, we performed a comprehensive study to analyze all research performed in this area by classifying established topics, identifying trends, and outlining areas for future research. Our three main contributions are as follows:

- We provide an extensive taxonomy of FoS in NLP research shown in Figure 2.
- We systematically classify research papers included in the ACL anthology and report findings on the development of FoS in NLP.
- We identify trends in NLP research and highlight directions for future work.

Our study highlights the development and current state of NLP research. Although we cannot fully cover all relevant work on this topic, we aim to provide a representative overview that can serve as a starting point for both NLP scholars and practitioners. In addition, our analysis can assist the research community bridge existing gaps and explore various FoS in NLP.

2 Related Work

Related literature that considers various different FoS in NLP is relatively scarce. Most studies focus only on a particular FoS or sub-field of NLP research.

For example, related studies focus on knowledge graphs in NLP (Schneider et al., 2022), explainability in NLP (Danilevsky et al., 2020), ethics and biases in NLP (Šuster et al., 2017; Blodgett et al., 2020), question answering (Liu et al., 2022b),

or knowledge representations in language models (Safavi and Koutra, 2021).

Studies that analyze NLP research based on the entire ACL Anthology focus on citation analyses (Mohammad, 2020a; Rungta et al., 2022) or visualizations of venues, authors, and n-grams and keywords extracted from publications (Mohammad, 2020b; Parmar et al., 2020).

Anderson et al. (2012) apply topic modeling to identify different epochs in the ACL’s history.

Various books categorize different FoS in NLP, focusing on detailed explanations for each of these categories (Allen, 1995; Manning and Schütze, 1999; Jurafsky and Martin, 2009; Eisenstein, 2019; Tunstall et al., 2022).

This study classifies and analyzes publications based on a broad spectrum of different FoS in NLP research to identify trends and possible directions for future work.

3 Research Questions

The goal of our study is an extensive analysis of research performed in NLP by classifying established topics, identifying trends, and outlining areas for future research. These objectives are reflected in our research questions (RQs) presented as follows:

RQ1: *What are the different FoS investigated in NLP research?*

Although most FoS in NLP are well-known and defined, there currently exists no commonly used taxonomy or categorization scheme that attempts to collect and structure these FoS in a consistent and understandable format. Therefore, getting an overview of the entire field of NLP research is difficult, especially for students and early career researchers. While there are lists of NLP topics in

conferences and textbooks, they tend to vary considerably and are often either too broad or too specialized. To classify and analyze developments in NLP, we need a taxonomy that encompasses a wide range of different FoS in NLP. Although this taxonomy may not include all possible NLP concepts, it needs to cover a wide range of the most popular FoS, whereby missing FoS may be considered as subtopics of the included FoS. This taxonomy serves as an overarching classification scheme in which NLP publications can be classified according to at least one of the included FoS, even if they do not directly address one of the FoS, but only subtopics thereof.

RQ2: *How to classify research publications according to the identified FoS in NLP?*

Classifying publications according to the identified FoS in NLP is very tedious and time-consuming. Especially with a large number of FoS and publications, a manual approach is very costly. Therefore, we need an approach that can automatically classify publications according to the different FoS in NLP.

RQ3: *What are the characteristics and developments over time of the research literature in NLP?*

To understand past developments in NLP research, we examine the evolution of popular FoS over time. This will allow a better understanding of current developments and help contextualize them.

RQ4: *What are the current trends and directions of future work in NLP research?*

Analyzing the classified research publications allows us to identify current research trends and gaps and predict possible future developments in NLP research.

4 Classification & Analysis

In this section, we report the approaches and results of the data classification and analysis. It is structured according to the formulated RQs.

4.1 Taxonomy of FoS in NLP research (RQ1)

To develop the taxonomy of FoS in NLP shown in Figure 2, we first examined the submission topics of recent years as listed on the websites of major NLP conferences such as ACL, EMNLP, COLING, or IJCNLP. In addition, we reviewed the topics of workshops included in the ACL Anthology to derive further FoS. In order to include smaller topics that are not necessarily mentioned on conference or workshop websites, we manually reviewed all papers from the recently published EMNLP 2022 Proceedings, extracted their FoS, and annotated all 828 papers accordingly. This provided us with an initial set of FoS, which we used to create the first version of the NLP taxonomy.

Based on our initial taxonomy, we conducted semi-structured expert interviews with NLP researchers to evaluate and adjust the taxonomy. In the interviews, we placed particular emphasis on the evaluation of the mapping of lower-level FoS to their higher-level FoS, and the correctness and completeness of FoS in the NLP domain. In total, we conducted more than 20 one-on-one interviews with different domain experts. After conducting the interviews, we noticed that experts demonstrated a high degree of agreement on certain aspects of evaluation, while opinions were highly divergent on other aspects. While we easily implemented changes resulting based on high expert agreement, we acted as the final authority in deciding whether to implement a particular change for aspects with low expert agreement. For example, one of the aspects with the highest agreement was that certain

Dataset →	Validation			Test		
Model ↓	P	R	F ₁	P	R	F ₁
BERT	96.57±0.14	95.43±0.16	96.00±0.03	89.77±0.20	93.58±0.07	91.64±0.10
RoBERTa	95.77±0.19	95.19±0.16	95.48±0.17	87.46±2.75	93.29±0.10	90.27±1.42
SciBERT	96.44±0.17	95.65±0.14	96.05±0.10	90.18±3.17	94.05±0.06	92.06±1.65
SPECTER 2.0	96.44±0.11	95.69±0.14	96.06±0.08	92.46±2.58	93.99±0.22	93.21±1.39
SciNCL	96.39±0.11	95.71±0.09	96.05±0.04	89.97±1.85	93.74±0.18	91.81±0.93

Table 1: Evaluation results for classifying papers according to the NLP taxonomy on three runs over different random train/validation splits. Since the distribution of classes is very unbalanced, we report micro scores.

Field of Study	# Papers	Representative Papers	Field of Study	# Papers	Representative Papers
Machine Translation	12,922	Liu et al. (2020), Goyal et al. (2022)	Visual Data in NLP	2,401	Tan and Bansal (2019), Xu et al. (2021)
Language Models	11,005	Devlin et al. (2019), Ouyang et al. (2022)	Ethical NLP	2,322	Blodgett et al. (2020), Perez et al. (2022)
Representation Learning	6,370	Reimers and Gurevych (2019), Gao et al. (2021b)	Question Answering	2,208	Karpukhin et al. (2020), Liu et al. (2022b))
Text Classification	6,117	Wei and Zou (2019), Hu et al. (2022)	Tagging	1,968	Malmi et al. (2019), Wei et al. (2020)
Low-Resource NLP	5,863	Gao et al. (2021a), Liu et al. (2022a)	Summarization	1,856	Liu and Lapata (2019), He et al. (2022)
Dialogue Systems & Conversational Agents	4,678	Zhang et al. (2020), Roller et al. (2021)	Green & Sustainable NLP	1,780	Strubell et al. (2019), Ben Zaken et al. (2022)
Syntactic Parsing	4,028	Zhou and Zhao (2019), Glavaš and Vulić (2021)	Cross-Lingual Transfer	1,749	Conneau et al. (2020), Feng et al. (2022)
Speech & Audio in NLP	3,915	Baevski et al. (2022), Wang et al. (2020)	Morphology	1,749	McCarthy et al. (2020), Goldman et al. (2022)
Knowledge Representation	2,967	Schneider et al. (2022), Safavi and Koutra (2021)	Explainability & Interpretability in NLP	1,671	Danilevsky et al. (2020), Pruthi et al. (2022)
Structured Data in NLP	2,803	Herzig et al. (2020), Yin et al. (2020)	Robustness in NLP	1,621	Hendrycks et al. (2020), Meade et al. (2022)

Table 2: Overview of the most popular FoS in NLP literature. Representative papers consist of either highly cited studies or comprehensive surveys on the respective FoS.

lower-level FoS must be assigned not only to one but also to multiple higher-level FoS.

Based on the interview results, we subsequently adjusted the annotations of the 828 EMNLP 2022 papers and developed the final NLP-taxonomy, as shown in Figure 2.

4.2 Field of Study Classification (RQ2)

We trained a weakly supervised classifier to classify ACL Anthology papers according to the NLP taxonomy. To obtain a training dataset, we first defined keywords for each FoS included in the final taxonomy to perform a database search for relevant articles. Based on the keywords, we created search strings to query the Scopus and arXiv databases. The search string was applied to titles and author keywords, if available. While we limited the Scopus search results to the NLP domain with additional restrictive keywords such as "NLP", "natural language processing", or "computational linguistics", we limited the search in arXiv to the cs.CL domain. We subsequently merged duplicate articles to create a multi-label dataset and removed articles included in the EMNLP 2022 proceedings, as this dataset is used as test set. Finally, we applied a fuzzy string matching heuristic and added missing classes based on the previously defined FoS keywords that appear twice or more in the article titles or abstracts. The final training dataset consists of 178,521 articles annotated on average with 3.65 different FoS. On average, each class includes 7936.50 articles, while the most frequent

class is represented by 63728 articles and the least frequent class by 141 articles. We split this unevenly distributed dataset into three different random 90/10 training/validation sets and used the human-annotated EMNLP 2022 articles as the test dataset.

For multi-label classification, BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), SciBERT (Beltagy et al., 2019), SPECTER 2.0 (Cohan et al., 2020; Singh et al., 2022), and SciNCL (Ostendorff et al., 2022) models were fine-tuned in their base versions on the three different training datasets and evaluated on their respective validation and test datasets. We train all models for three epochs, using a batch size of 8, a learning rate of $5e-5$, and the AdamW optimizer (Loshchilov and Hutter, 2019).

The evaluation results are shown in Table 1. SPECTER 2.0 shows significant performance on both validation and test data. Therefore, we selected SPECTER 2.0 as our final classification model, which we subsequently trained with the same parameters on the combined training, validation, and test data. Using this final model, we classified all papers included in the ACL Anthology from 1952 to 2022. To obtain our final dataset for analysis, we removed the articles that were not truly research articles, such as prefaces; articles that were not written in English; and articles where the classifier was uncertain and simply predicted every class possible. This final classified dataset includes a total of 74,279 research papers.

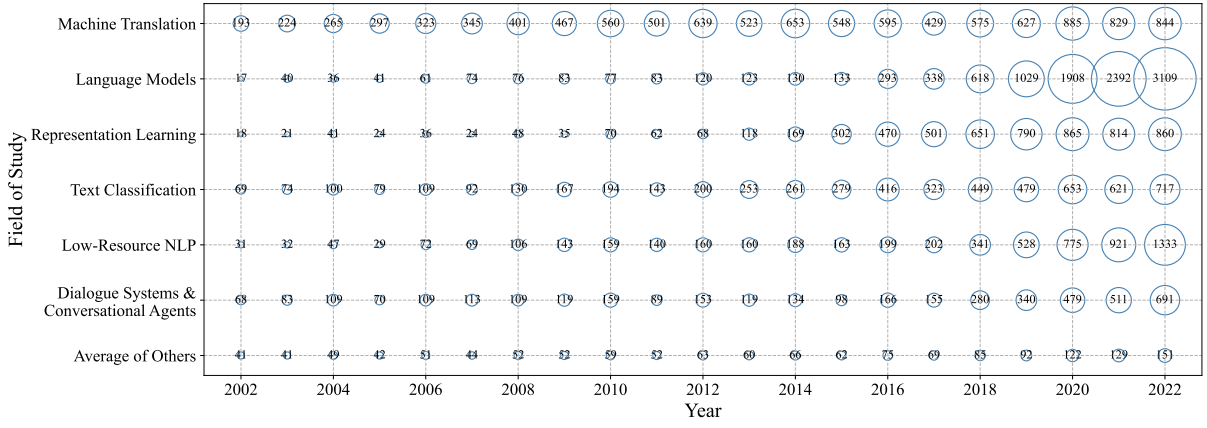


Figure 3: Distribution of number of papers by most popular FoS from 2002 to 2022.

Table 2 shows the final classification results with respect to the number of publications for each of the most popular FoS.

4.3 Characteristics and Developments of the Research Landscape (RQ3)

Considering the literature on NLP, we start our analysis with the number of studies as an indicator of research interest. The distribution of publications over the 50-year observation period is shown in Figure 1. While the first publications appeared in 1952, the number of annual publications grew slowly until 2000. Accordingly, between 2000 and 2017, the number of publications roughly quadrupled, whereas in the subsequent five years it has doubled again. We therefore observe a near-exponential growth in the number of NLP studies, indicating increasing attention from the research community.

Examining Table 2 and Figure 3, the most popular FoS in the NLP literature and their recent development over time are revealed. While the majority of studies in NLP are related to machine translation or language models, the developments of both FoS are different. Machine translation is a thoroughly researched field that has been established for a long time and has experienced a modest growth rate over the last 20 years. Language models have also been researched for a long time. However, the number of publications on this topic has only experienced significant growth since 2018. Similar differences can be observed when looking at the other popular FoS. Representation learning and text classification, while generally widely researched, are partially stagnant in their growth. In contrast, dialogue systems & conversational agents and particularly

low-resource NLP continue to exhibit high growth rates in the number of studies. Based on the development of the average number of studies on the remaining FoS in Figure 3, we observe a slightly positive growth overall. However, the majority of FoS are significantly less researched than the most popular FoS. We conclude that the distribution of research across FoS is extremely unbalanced and that the development of NLP research is largely shaped by advances in a few highly popular FoS.

4.4 Research Trends and Directions for Future Work (RQ4)

Figure 4 shows the growth-share matrix of FoS in NLP research inspired by Henderson (1970). We use it to examine current research trends and possible future research directions by analyzing the growth rates and total number of papers related to the various FoS in NLP between 2018 and 2022. The upper right section of the matrix consists of FoS that exhibit a high growth rate and simultaneously a large number of papers overall. Given the growing popularity of FoS in this section, we categorize them as *trending stars*. The lower right section contains FoS that are very popular but exhibit a low growth rate. Usually these are FoS that are essential for NLP research but already relatively mature. Hence, we categorize them as *foundational FoS*. The upper left section of the matrix contains FoS that exhibit a high growth rate, but only very few papers overall. Since the progress of these FoS is rather promising, but the small number of overall papers renders it difficult to predict their further developments, we categorize them as *rising question marks*. The FoS in the lower left of the matrix are categorized as *niche FoS* owing to their low total

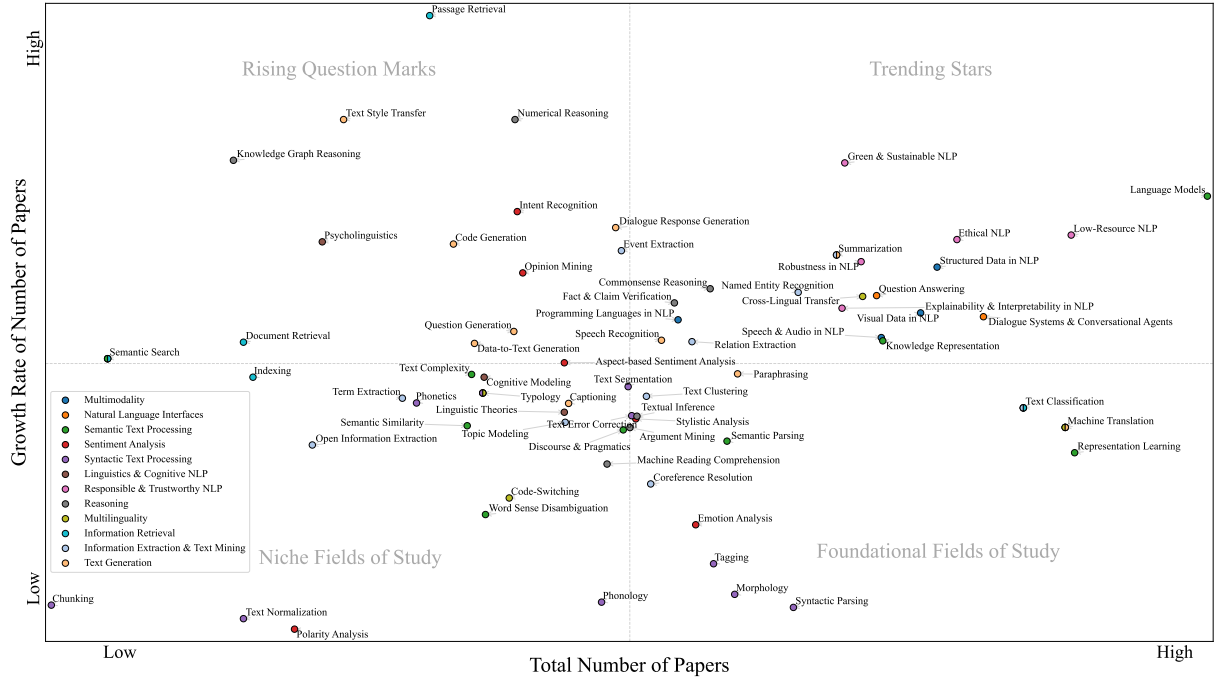


Figure 4: Growth-share matrix of FoS in NLP. The growth rates and total number of works for each FoS are calculated from the start of 2018 to the end of 2022. To obtain a more uniform distribution of the data, we apply the Yeo-Johnson transformation (Yeo and Johnson, 2000).

number of papers and their low growth rates.

Figure 4 shows that language models are currently receiving the most attention, which is also consistent with the observations from Table 2 and Figure 3. Based on the latest developments in this area, this trend is likely to continue and accelerate in the near future. Text classification, machine translation, and representation learning rank among the most popular FoS, but only show marginal growth. In the long term, they may be replaced by faster-growing fields as the most popular FoS.

In general, FoS related to syntactic text processing exhibit negligible growth and low popularity overall. Conversely, FoS concerned with responsible & trustworthy NLP, such as green & sustainable NLP, low-resource NLP, and ethical NLP tend to exhibit a high growth rate and also high popularity overall. This trend can also be observed in the case of structured data in NLP, visual data in NLP, and speech & audio in NLP, all of which are concerned with multimodality. In addition, natural language interfaces involving dialogue systems & conversational agents and question answering are becoming increasingly important in the research community. We conclude that in addition to language models, responsible & trustworthy NLP, multimodality, and natural language interfaces are likely to characterize the NLP research landscape

in the near future.

Further notable developments can be observed in the area of reasoning, specifically with respect to knowledge graph reasoning and numerical reasoning and in various FoS related to text generation. Although these FoS are currently still relatively small, they apparently attract more and more interest from the research community and show a clear positive tendency toward growth.

Figure 5 shows the innovation life cycle of the most popular FoS in NLP adapted from the *diffusion of innovations* theory (Rogers, 1962) and inspired by Huber (2005). The central assumption of the innovation life cycle theory is that for each innovation (or in this case FoS), the number of published research per year is normally distributed over time, while the total number of published research reaches saturation according to a sigmoid curve.

We consider the following aspects which influence the position of a FoS on the innovation life cycle curve:

- A high growth rate in the number of publications indicates that FoS are at the beginning of their life cycle, while a stagnant or negative growth rate indicates a tendency toward maturation.
- If the number of recently published papers ac-

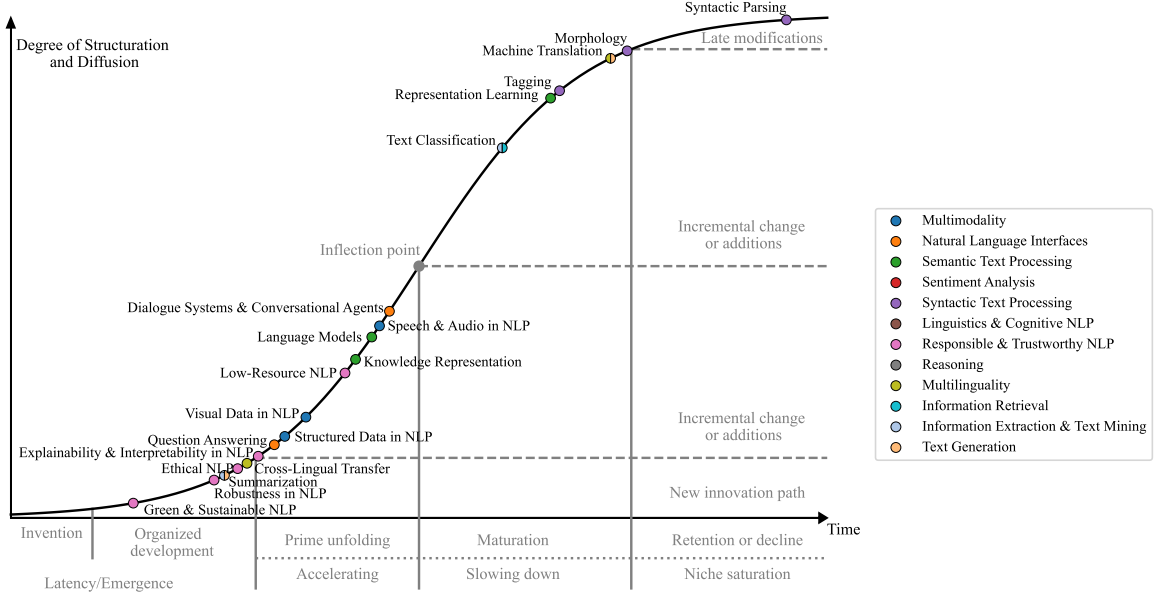


Figure 5: Innovation life cycle of the most popular FoS in NLP. FoS on the left side of the curve are at the beginning of their life cycle. They have just been invented or are in an early phase, where innovation on FoS accelerates by a rising number of studies. After passing the inflection point, the FoS move towards the end of their innovation life cycle, where research on FoS is retained, or declines, and only late modifications are added to the FoS.

counts for a significant percentage of the total number of papers published on a certain FoS over time, this indicates the new development of a FoS.

- A high percentage of recent publications on a certain FoS compared to the total number of recent publications on all FoS indicates the maturity of a FoS and its adoption by the majority of researchers.

Accordingly, we define the position of a set of FoS $F = \{f_1, \dots, f_m\}$ on the x-axis of the innovation life cycle curve as:

$$x_{f_i} = \log\left(\frac{1}{g_{t-n,t}(f_i)} \cdot \frac{1}{h_{t-n,t}(f_i)} \cdot k_{t-n,t}(f_i)\right) \quad (1)$$

where t is a specific year, $g_{t-n,t}(f_i)$ is the growth rate of the number of publications for a particular FoS, normalized between $1e^{-10}$ and 1, $h_{t-n,t}(f_i)$ is the percentage of the number of papers for a particular FoS in a chosen time period compared to the total number of papers for the same FoS over the entire observed time period, and $k_{t-n,t}(f_i)$ is the percentage of the number of publications for a specific FoS in a chosen time period compared to all publications in the same time period across all FoS. We choose a five-year time period with

$t = 2022$ and $n = 5$, while the entire observed time period ranges from 1952 to 2022. Finally, to map the FoS to the innovation life cycle curve, we normalize $X = \{x_{f_1}, \dots, x_{f_m}\}$ between lower and upper bounds of -5 and 5 as $X' = \{x'_{f_1}, \dots, x'_{f_m}\}$ and calculate their position on the y-axis of the innovation life cycle curve as:

$$y_{f_i} = \frac{1}{1 + e^{-x'_{f_i}}}. \quad (2)$$

From Figure 5, we observe that FoS related to syntactic text processing are already relatively mature and approaching the end of the innovation life cycle. Particularly syntactic parsing is getting near the end of its life cycle, with only late modifications being researched. While Table 2 shows that machine translation, representation learning, and text classification are very popular overall, Figure 5 reveals that they have passed the inflection point of the innovation life cycle curve and their development is currently slowing down. They are adopted by most researchers but show stagnant or negative growth, as also indicated in Figure 4. However, most of FoS have not yet reached the inflection point and are still experiencing increasing growth rates, while research on these FoS is accelerating. Especially FoS related to responsible & trustworthy NLP, multimodality, and natural language inter-

faces are just beginning their innovation life cycle, suggesting that research in these areas will likely accelerate in the following years. This is also in line with the observations from Figure 4, where most of the FoS related to these areas are categorized as *trending stars*. Further, we observe that language models have passed the first two stages of innovation and are currently in their prime unfolding phase. They are adopted by a large number of researchers and research on them is still accelerating. Comparing this to Figure 4, where language models are among the most trending FoS, we conclude that this trend is likely to continue in the near future and is unlikely to slow down anytime soon.

5 Discussion

The observations of our comprehensive study reveal several insights that we can situate to related work. Since the first publications in 1952, researchers have paid increasing attention to the field of NLP, particularly after the introduction of Word2Vec (Mikolov et al., 2013) and accelerated by BERT (Devlin et al., 2019). This observed growth in research interest is in line with the study of Mohammad (2020b). Historically, machine translation was one of the first research fields in NLP (Jones, 1994), which continues to be popular and steadily growing nowadays. However, recent advances in language model training have sparked increasing research efforts in this field, as shown in Figure 3. Since scaling up language models enhance performance on downstream tasks (Brown et al., 2020; Kaplan et al., 2020; Wei et al., 2022a; Hoffmann et al., 2022), researchers continue to introduce increasingly larger language models (Han et al., 2021). However, training and using these large language models involves significant challenges, including computational costs (Narayanan et al., 2021), environmental issues (Strubell et al., 2019), and ethical considerations (Perez et al., 2022). As a result, a recent increase in research efforts has been noted to render language models and NLP more responsible & trustworthy in general, as shown in Figure 4 and Figure 5. Additionally, recent advances aim to train large-scale multimodal language models capable of understanding and generating natural language text and performing all types of downstream tasks while interacting with humans through natural language input prompts. (OpenAI, 2023). From our observations in Figure 4 and Figure 5 we again find support for this trend in

NLP literature for multimodality, text generation, and natural language interfaces.

Although language models have achieved remarkable success on various NLP tasks, their inability to reason is often seen as a limitation that cannot be overcome by increasing the model size alone (Rae et al., 2022; Wei et al., 2022b; Wang et al., 2023). Although reasoning capabilities are a crucial prerequisite for the reliability of language models, this field is still relatively less researched and receives negligible attention. While Figure 4 exhibits high growth rates for knowledge graph reasoning and numerical reasoning in particular, research related to reasoning is still rather under-represented compared to the more popular FoS.

6 Conclusion

Recent years have witnessed an increasing prominence of NLP research. To summarize recent developments and provide an overview of this research area, we defined a taxonomy of FoS in NLP and analyzed recent research developments.

Our findings show that a large number of FoS have been studied, including trending fields such as multimodality, responsible & trustworthy NLP, and natural language interfaces. While recent developments are largely a result of recent advances in language models, we have noted a lack of research pertaining to teaching these language models to reason and thereby afford more reliable predictions.

7 Limitations

Constructing the taxonomy highly depends on the personal decisions of the authors, which can bias the final result. The taxonomy may not cover all possible FoS and offers potential for discussions, as domain experts have inherently different opinions. As a countermeasure, we aligned the opinions of multiple domain experts and designed the taxonomy at a higher level, allowing non-included FoS to be considered as possible subtopics of existing ones.

For this study, we limited our analysis to papers published in the ACL Anthology, which typically feature research presented at major international conferences and are written in English. However, research communities that publish their work in regional venues exist, often in languages other than English. In addition, NLP research is also presented at other prominent global conferences such as AAAI, NeurIPS, ICLR, or ICML. Therefore, the

findings we report in this study pertain specifically to NLP research presented at major international conferences and journals in English.

Furthermore, the accuracy of the classification results poses another threat to the validity of our study. Data extraction bias and classification model errors may negatively affect the results. To mitigate this risk, the authors regularly discussed the used classification schemes and conducted a thorough evaluation of the performance of the classification model.

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A Appendix

A.1 Descriptions of fields-of-studies included in the NLP taxonomy

In the following, we explain the fields-of-study concepts included in Figure 2.

A.1.1 Multimodality

Multimodality refers to the capability of a system or method to process input of different types or “modalities” (Garg et al., 2022). We distinguish between systems that can process text in natural language along with **visual data**, **speech & audio**, **programming languages** or **structured data** such as tables or graphs.

A.1.2 Natural Language Interfaces

Natural language interfaces can process data based on natural language queries (Voigt et al., 2021), usually implemented as **question answering** or **dialogue & conversational systems**.

A.1.3 Semantic Text Processing

This high-level FoS includes all types of concepts that attempt to derive meaning from natural language and enable machines to interpret textual data semantically. One of the most powerful FoS in this regard are **language models** that attempt to learn the joint probability function of sequences of words (Bengio et al., 2000). Recent advances in language model training have enabled these models to successfully perform various downstream NLP tasks (Soni et al., 2022). In **representation learning**, semantic text representations are usually learned in the form of embeddings (Fu et al., 2022; Schopf et al., 2023d,c), which can be used to compare the **semantic similarity** of texts in **semantic search** settings (Reimers and Gurevych, 2019). Additionally, **knowledge representations**, e.g., in the form of knowledge graphs, can be incorporated to improve various NLP tasks (Schneider et al., 2022).

A.1.4 Sentiment Analysis

Sentiment analysis attempts to identify and extract subjective information from texts (Wankhade et al., 2022). Usually, studies focus on extracting **opinions**, **emotions**, or **polarity** from texts. More recently, **aspect-based sentiment analysis** emerged as a way to provide more detailed information than general sentiment analysis, as it aims to predict the sentiment polarities of given aspects or entities in text (Xue and Li, 2018).

A.1.5 Syntactic Text Processing

This high-level FoS aims at analyzing the grammatical syntax and vocabulary of texts (Bessmertny et al., 2016). Representative tasks in this context are **syntactic parsing** of word dependencies in sentences, **tagging** of words to their respective part-of-speech, **segmentation** of texts into coherent sections, or **correction of erroneous texts** with respect to grammar and spelling.

A.1.6 Linguistics & Cognitive NLP

Linguistics & Cognitive NLP deals with natural language based on the assumptions that our linguistic abilities are firmly rooted in our cognitive abilities, that meaning is essentially conceptualization, and that grammar is shaped by usage (Dabrowska and Divjak, 2015). Many different **linguistic theories** are present that generally argue that language acquisition is governed by universal grammatical rules that are common to all typically developing humans (Wise and Sevcik, 2017). **Psycholinguistics** attempts to model how a human brain acquires and produces language, processes it, comprehends it, and provides feedback (Balamurugan, 2018). **Cognitive modeling** is concerned with modeling and simulating human cognitive processes in various forms, particularly in a computational or mathematical form (Sun, 2020).

A.1.7 Responsible & Trustworthy NLP

Responsible & trustworthy NLP is concerned with implementing methods that focus on fairness, **explainability**, accountability, and **ethical** aspects at its core (Barredo Arrieta et al., 2020). **Green & sustainable NLP** is mainly focused on efficient approaches for text processing, while **low-resource NLP** aims to perform NLP tasks when data is scarce. Additionally, **robustness in NLP** attempts to develop models that are insensitive to biases, resistant to data perturbations, and reliable for out-of-distribution predictions.

A.1.8 Reasoning

Reasoning enables machines to draw logical conclusions and derive new knowledge based on the information available to them, using techniques such as deduction and induction. **Argument mining** automatically identifies and extracts the structure of inference and reasoning expressed as arguments presented in natural language texts (Lawrence and Reed, 2019). **Textual inference**, usually modeled as entailment problem, automatically determines

whether a natural-language *hypothesis* can be inferred from a given *premise* (MacCartney and Manning, 2007). **Commonsense reasoning** bridges premises and hypotheses using world knowledge that is not explicitly provided in the text (Ponti et al., 2020), while **numerical reasoning** performs arithmetic operations (Al-Negheimish et al., 2021). **Machine reading comprehension** aims to teach machines to determine the correct answers to questions based on a given passage (Zhang et al., 2021).

A.1.9 Multilinguality

Multilinguality tackles all types of NLP tasks that involve more than one natural language and is conventionally studied in **machine translation**. Additionally, **code-switching** freely interchanges multiple languages within a single sentence or between sentences (Diwan et al., 2021), while **cross-lingual transfer** techniques use data and models available for one language to solve NLP tasks in another language.

A.1.10 Information Retrieval

Information retrieval is concerned with finding texts that satisfy an information need from within large collections (Manning et al., 2008). Typically, this involves retrieving **documents** or **passages**.

A.1.11 Information Extraction & Text Mining

This FoS focuses on extracting structured knowledge from unstructured text and enables the analysis and identification of patterns or correlations in data (Hassani et al., 2020). **Text classification** automatically categorizes texts into predefined classes Schopf et al. (2021, 2023a,b), while **topic modeling** aims to discover latent topics in document collections (Grootendorst, 2022), often using **text clustering** techniques that organize semantically similar texts into the same clusters. **Summarization** produces summaries of texts that include the key points of the input in less space and to keep repetition to a minimum (El-Kassas et al., 2021). Additionally, the information extraction & text mining FoS also includes **named entity recognition**, which deals with the identification and categorization of named entities, (Leitner et al., 2020), **coreference resolution** that aims to identify all references to the same entity in discourse (Yin et al., 2021), **term extraction** that aims to extract relevant terms such as keywords or keyphrases (Rigouts Terryn et al., 2020; Schopf et al., 2022), **relation extraction** that aims to extract relations

between entities, and **open information extraction** that facilitates the domain-independent discovery of relational tuples (Yates et al., 2007).

A.1.12 Text Generation

The objective of text generation approaches is to generate texts that are both comprehensible to humans and indistinguishable from text authored by humans. Accordingly, the input usually consists of text, such as in **paraphrasing** that renders the text input in a different surface form while preserving the semantics (Niu et al., 2021), **question generation** that aims to generate a fluid and relevant question given a passage and a target answer (Song et al., 2018), or **dialogue-response generation** which aims to generate natural-looking text relevant to the prompt (Zhang et al., 2020). In many cases, however, the text is generated as a result of input from other modalities, such as in the case of **data-to-text generation** that generates text based on structured data such as tables or graphs (Kale and Rastogi, 2020), **captioning** of images or videos, or **speech recognition** that transcribes a speech waveform into text (Baevski et al., 2022).